24 October 2026

The Editor
Journal of Flood Risk Management

Dear Editor,

I am pleased to submit the manuscript “When do flood susceptibility models beat the flood record? Evidence from radar-observed floods in Bangladesh” for consideration as an Original Paper in the Journal of Flood Risk Management.

Every flood manager in a radar-mapped region already holds a free flood map, the record of where floods have been, yet published flood susceptibility maps are almost never tested against it. The manuscript sets a gradient-boosted terrain model against that record on later floods mapped from Sentinel-1 radar, scoring both on 10 km blocks held out from training, and asks when the record is enough and when a model adds to it.

Across 15 tests in seven regions, the record’s lead fell as the share of a flood reaching ground no earlier flood had reached rose (Spearman $\rho = -0.775$, $p < 0.001$), and the two were level at about a third of flooded assets on new ground. The relation held with each region left out in turn. Where floods returned to the same ground the record won (AUC 0.896 against 0.801 in Rangpur and Rajshahi), and on the cyclone-affected coast the model led. A global hydrodynamic hazard map was no stronger than the model.

The paper suits the Journal because its conclusions concern how flood risk is managed. It recommends that agencies publish the radar flood record as a planning baseline, and that the choice between record and model follow how much of past flooding fell on new ground. Its limits are stated plainly: scores measure agreement with flooding as radar sees it, 10 of the 15 tests cannot separate record from model, and all regions lie in one country. The code is at https://github.com/rakibhhridoy/rimes-mapathon and the data at https://doi.org/10.5281/zenodo.23573843.

The main text runs to just under 6,000 words with 5 figures and 7 tables, plus a Supporting Information file. The manuscript is original and not under consideration elsewhere. The software it uses is described in a separate manuscript under review at SoftwareX (SOFTX-D-26-01375), which reports the software rather than these findings. I am the sole author, declare an affiliation with Fermium Systems, which hosts the public dashboard, and received no specific funding.

Suggested reviewers:

- Dr Beth Tellman, Nelson Institute for Environmental Studies, University of Wisconsin–Madison, USA. Beth.tellman@wisc.edu
- Dr Sandro Martinis, German Remote Sensing Data Center, German Aerospace Center (DLR), Germany. Sandro.Martinis@dlr.de
- Prof. Hanna Meyer, Institute of Landscape Ecology, University of Münster, Germany. hanna.meyer@uni-muenster.de
- Dr Alexandre Wadoux, James Cook University, Australia. alexandre.wadoux@yahoo.fr

Thank you for considering the manuscript.

Yours sincerely,

Md Rakib Hasan
Department of Soil, Water and Environment, University of Dhaka, Dhaka-1000, Bangladesh
Fermium Systems, Dhaka-1207, Bangladesh
rakibhhridoy@fermium.systems · ORCID 0009-0002-4007-7590